

GOV-AR: Risk-Bounded Multi-Tenant LLM Admission and Routing under Delayed Token-Cost Feedback

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Abstract—Enterprise LLM gateways increasingly share model endpoints across multiple tenants, yet the final output-token cost of an individual request is only known after completion and settlement may arrive late. This creates an online control problem that is not addressed by post-hoc FinOps reporting alone. We present GOV-AR, a reproducible implementation path for governance-aware admission, routing, and reservation under delayed token-cost feedback. GOV-AR combines a hard governance filter, a reservation policy over predicted output lengths, tenant-isolated settled versus reserved budget accounting, and first-class `admit`, `queue`, `reject`, and `abstain` actions. The method is implemented as a research workspace plus an operator-side admission service aligned with an existing Kubernetes operator for AI FinOps and sovereignty. We verify the implementation with trace-driven experiments covering delayed settlement, multi-tenant contention, drift response, fault injection, scalability, live Azure validation, and component ablation. In the current evidence, quantile reservation achieves higher throughput than mean-plus-standard-deviation reservation in one delayed-settlement trace, while mean-plus-standard-deviation avoids overshoot in other cases at the cost of higher slack. We also validate a local Docker/Kind/Helm execution path and live Azure OpenAI and Foundry invocations using synthetic prompts. The result is a reproducible experimental basis for a fuller operator-integrated GOV-AR system, including an optional PostgreSQL-backed ledger and a proxy-triggered admit/settle/cancel path, with explicit documentation of remaining gaps for literature hardening, deeper Envoy-native integration, and final publication packaging.

Index Terms—LLM gateways, admission control, delayed feedback, Kubernetes operators, multi-tenant systems, FinOps, sovereign AI, Azure OpenAI.



1 INTRODUCTION

Large-language-model gateways now act as shared infrastructure for multiple applications, teams, and tenants. In regulated enterprises, each routing decision must respect cost ceilings, provider and residency constraints, and application-specific guardrails. However, the output token count of a request is unknown at admission time, multiple requests can be in flight concurrently, and cost telemetry may arrive only after delayed settlement. These properties make budget safety an online systems problem rather than a purely retrospective reporting problem.

The operator in this repository already provides telemetry ingestion, cost attribution, budget phase tracking, sovereignty enforcement, and route recommendation. What it lacks is a first-class reserve-dispatch-settle method for uncertain, concurrent, and delayed-cost requests. GOV-AR addresses that gap.

1.0.0.1 Contributions. This manuscript makes three scoped contributions consistent with the current state of implementation:

- 1) a formulation of multi-tenant LLM admission and routing under delayed token-cost feedback with settled and reserved budget views;
- 2) a reproducible GOV-AR scaffold with governance filtering, reservation, idempotent settlement, queueing, abstention, and delayed-settlement replay;
- 3) an evidence package spanning E0–E7 experiments, local Docker/Kind/Helm execution, and live Azure

OpenAI plus Foundry validation with synthetic prompts.

2 RELATED WORK

Adaptive LLM routing under budget constraints has been studied in recent work such as Adaptive LLM Routing under Budget Constraints [1]. Cost-aware cascades and routers such as FrugalGPT [2], RouteLLM [3], and HYBRID LLM [4] show that query-dependent model selection can materially improve cost-quality trade-offs. Recent conformal routing work [5] adds distribution-free safety guarantees to the routing decision itself. Recent ACL 2026 work further expands the routing design space: InferenceDynamics targets adaptable routing across broad model pools, FLARE emphasizes length-aware resource efficiency, and MTRouter extends cost-aware routing to multi-turn trajectories [6], [7], [8]. Delayed feedback and online resource-constrained decision-making also have deeper theoretical roots in online learning under delayed feedback [9] and contextual bandits with knapsack-style resource constraints [10], [11]. Output-length-aware serving further motivates prediction-based reservation [12], [13]. The gap targeted here is their combination with tenant-isolated budget liability, explicit queueing and abstention decisions, hard governance filtering, and a Kubernetes-native control-plane execution context. None of the verified primary sources above combines delayed settlement, in-flight financial liability, tenant-isolated

reservation accounting, and operator-integrated governance enforcement in one end-to-end method.

3 PROBLEM FORMULATION

For each incoming request r from tenant t , GOV-AR chooses one of `admit(model, reservation)`, `queue`, `reject`, or `abstain`. Hard governance constraints filter infeasible providers and models before soft scoring begins. At decision time, the system knows the tenant, request metadata, historical output-length samples, and current settled/reserved budget state, but not the final output token count nor the ultimate settled cost.

The key accounting abstraction is:

$$\text{available}_t(k) = \text{budget}_t - \text{settled}_t(k) - \text{reserved}_t(k).$$

This separates already committed spend from in-flight liability. A request can only be dispatched after its reservation is atomically committed.

4 GOV-AR DESIGN

The current GOV-AR scaffold has four core pieces.

4.0.0.1 Governance filtering.: Candidates that violate hard policy are removed before budget or utility are considered.

4.0.0.2 Reservation policy.: The scaffold currently supports empirical quantile reservation and mean plus standard deviation reservation over predicted output token samples.

4.0.0.3 Tenant ledger.: Each tenant tracks both settled and reserved cost. A per-request ledger stores reservation amount, model, status, and settlement event identifier. Settlement is idempotent for repeated delivery of the same event and rejects conflicting duplicate events.

4.0.0.4 Replay engine.: The replay harness supports delayed settlement, queue retries, drift-aware abstention, and request-level event traces.

5 IMPLEMENTATION

The implementation is intentionally split between an operator audit of the existing platform and a research workspace in `article3/`. The workspace contains:

- `src/admission`: the decision layer with `admit`, `queue`, `reject`, and `abstain`;
- `src/ledger`: tenant budget state plus per-request reservation records and idempotent settlement;
- `src/predictor`: reservation bounds and simple drift detection;
- `src/trace_gateway`: trace generation, replay, and metrics;
- `cmd/experiment`: runners for E0–E7;
- `infra/`: local Docker, Kind, Helm, and Azure live-validation automation.

On the operator side, the implementation now also includes an optional `gov-ar-admission` service that reads `AIBudgetPolicy`, `AIRoutingPolicy`, `AIModel`, and `AIPProvider`, exposes `/v1/admit`, `/v1/settle`, `/v1/cancel`, and `/v1/liability/{tenant}`, and can

persist reservations and settlements either in memory or in PostgreSQL. A proxy-triggered live path in the existing `header-proxy` can call `admit` before upstream dispatch and `settle` or `cancel` after response completion. The local infrastructure now includes dedicated experiment and admission-service container images, a Helm chart that runs the experiment binary as a Kubernetes Job, and Kind automation that successfully deployed and executed the job in a dedicated local cluster.

6 EVALUATION

The present evaluation is a reproducible evidence package rather than a final publication-grade study. Still, every numeric claim in this section is tied to a generated artifact under `article3/experiments/`.

6.1 E1 and E2: Delayed Settlement and Multi-Tenancy

In the deterministic E1 delayed-settlement trace, quantile reservation admitted two requests and settled a total cost of 12.0 with 2.0 slack and no overshoot. The mean-plus-standard-deviation variant admitted one request, queued one, and settled 7.0 with 1.0 slack and no overshoot. In the deterministic E2 multi-tenant trace, both variants admitted three requests and queued one, but the quantile variant incurred 0.5 overshoot while mean-plus-standard-deviation used more slack (1.5) and avoided overshoot.

6.2 E3 and E4: Drift and Faults

E3 demonstrates conservative fallback under drift. With a mean ratio drift of 2.08 over the configured threshold, the scaffold admitted a non-drifted fallback candidate for one request and abstained on a second request when no safe candidate remained. E4 validates that the ledger accepts the first settlement event, tolerates replay of the same event idempotently, rejects a conflicting second event, releases reservation on expiry, and abstains when the decision layer has insufficient fresh evidence.

6.3 E5: Scalability

The E5 scalability sweep replays synthetic traces from 50 to 1000 requests. The largest current run completes in sub-millisecond to millisecond-scale local replay time, with throughput in the hundreds of thousands of requests per second in this lightweight harness. These numbers characterize the replay engine, not an end-to-end production gateway.

6.4 E6: Azure Live Validation

E6 validates live infrastructure and real deployments already present in the subscription. The experiment discovered four cognitive-service resources and three relevant deployments: `gpt-france-mini` in `francecentral`, `gpt-us-mini` in `eastus`, and `mistral-large-latest` in a Foundry/AIServices account in `westus2`. Live calls with a synthetic prompt returned HTTP 200 and the expected text OK on both Azure OpenAI deployments and on the Foundry Mistral deployment, with token-usage and latency metadata captured in the raw artifact.

6.5 E7: Ablation

E7 compares quantile, mean-plus-standard-deviation, an aggressive low reservation baseline, and a no-queue policy. In the current budget-delay scenario, the aggressive low reservation variant yields higher settled total than conservative reservation but incurs overshoot, while disabling queueing turns deferred decisions into outright rejects. This supports the claim that both reservation policy and queue semantics materially shape the safety-throughput trade-off.

7 DISCUSSION

The strongest result of the current artifact is reproducibility across several axes at once: deterministic trace replay, live Azure validation, local containerization, cluster execution, and an operator-side admission service with a persistent ledger option. The weakest aspect is still depth of integration into the final operator control path. GOV-AR now includes a proxy-triggered reserve-settle path, but it is not yet an Envoy-native or controller-driven production integration. The literature review and manuscript also remain less mature than the implementation and experiment harness.

8 THREATS TO VALIDITY

The main threats are external validity and maturity bias. Synthetic traces do not capture the full burstiness or semantic variability of production prompts. Replay throughput does not equal production gateway throughput. Azure live validation used only short synthetic prompts and existing deployments; it does not by itself prove deployment-scale economics. Finally, the related-work corpus is materially stronger than in the initial draft, but the broader review still needs a final saturation pass before submission freeze.

9 CONCLUSION

GOV-AR frames enterprise LLM routing as a joint admission, reservation, and settlement problem under delayed token-cost feedback and hard governance constraints. The current artifact already demonstrates a reproducible decision kernel, a local Docker/Kind/Helm execution path, an operator-side admission service with optional PostgreSQL persistence, a proxy-triggered live admit/settle path, and successful Azure live validation against real Azure OpenAI and Foundry deployments. The next required steps are deeper Envoy-native operator integration, a final saturation pass on the literature review, and completion of the final paper and replication package.

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